

Research paper

Sustainable modernization of rural agricultural power systems in North Africa via optimal renewable microgrid design

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ABSTRACT

This paper, conducted within the framework of the MG-FARM project, presents an integrated methodology for the design, sizing, and experimental validation of a microgrid tailored to an isolated agricultural farm in rural Morocco. The proposed approach relies on a scaled-down functional prototype, built with certified industrial-grade equipment and tested under an accelerated-time protocol, enabling the dynamic simulation of a full agricultural day under realistic operating conditions. Energy sources were initially sized using PVSyst software (25 kWp - 2,828 Ah at 51 V). To select the architecture, a rigorous comparison between AC- and DC-coupling was conducted, factoring in specific constraints—namely, exclusively AC loads and an AC backup source typical of the Moroccan context. Based on developed mathematical efficiency models, this analysis demonstrated the AC configuration's superiority, yielding a 2.1 % efficiency gain. Building on this, a techno-economic optimization of the PV inverter identified an optimal sizing ratio tailored to local conditions, ensuring a balanced trade-off between energy performance and Levelized Cost of Energy. To justify the transition from the conventional architecture to the proposed microgrid system, an environmental and economic assessment revealed a 98 % reduction in CO₂ emissions and estimated savings of €26,000 compared to the reference configuration. Finally, experimental validation was performed using a reduced-scale AC microgrid configuration (2 kW - 120 Ah at 24 V), designed to meet rapid testing constraints. The prototype's design is based on a theoretical source sizing specifically adapted to the accelerated-time protocol. This validation confirmed the consistency of the proposed design and the continuity of electricity supply across various dynamic scenarios.

1. Introduction

Like many North African countries, Morocco remains heavily dependent on energy imports (≈ 90 % of primary supply) and fossil fuels, which still account for 88 % of the national energy mix [1]. According to [2], the national final electricity consumption reached 37.61 TWh in 2023, as illustrated in Fig. 1. In rural areas, where grid extension remains economically challenging, electrification serves as a significant driver for agricultural productivity and socio-economic development. Indeed, powering a typical community of 200 households requires > 400 kWh/day, a demand that can be met through a hybrid PV-wind-hydropower microgrid, featuring a levelized cost of energy (LCOE) of \$0.147/kWh and a Net Present Cost (NPC) of \$473.6k [1]. Although diesel-based approaches remain widespread in irrigation, they entail high operational costs and substantial emissions. Conversely, a

fully renewable solar/wind/battery system covering a daily load profile of 240 kWh reduces the LCOE to \$0.119/kWh, achieves an NPC of \$194k, and provides a usable energy surplus of 61.7 % [3]. This transition is further encouraged by Morocco's remarkable solar potential—the annual Global Horizontal Irradiation (GHI) ranges from 1,823 to 2,213 kWh m⁻² across six climatic zones, with average daily values peaking at 5.32 kWh m⁻² day⁻¹ at rural sites such as Ouenskra [1,4]. Moreover, integrating Distributed Energy Resources (DER) and electric vehicles, combined with optimization strategies such as Particle Swarm Optimization (PSO), has demonstrated up to a 42 % net cost reduction in pilot platforms such as the Green Energy Park [5]. These findings underscore the strategic potential of converting conventional hybrid installations into smart microgrids that are tailored to local climatic conditions and specific agricultural consumption profiles, thus supporting national objectives toward carbon neutrality and energy

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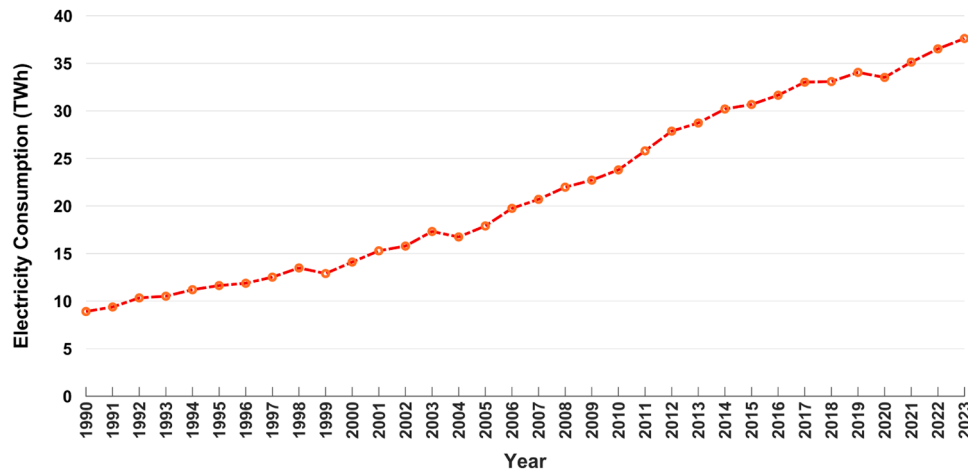


Fig. 1. Evolution of Morocco's National Electricity Consumption from 1990 to 2023.

independence.

Before large-scale deployment in agricultural, industrial, or residential contexts, hybrid microgrids must undergo rigorous experimental validations capable of verifying not only the consistency of theoretical models but also the functional reliability and operational suitability of systems under realistic conditions. Recent literature presents several experimental approaches primarily based on scaled-down prototypes designed using research-grade equipment or academic test benches [6–11]. Complementary studies have utilized Hardware-in-the-Loop (HIL) platforms to integrate dynamic real-time phenomena [12]. Although these studies provide valuable methodological insights, their direct transferability to practical agricultural environments remains limited, notably due to the specificity of the equipment used. Other research efforts have attempted to bridge this technological gap by incorporating certified industrial equipment, such as Studer Xtender XTM 4000–48 inverter-chargers, Sunny Boy photovoltaic inverters, or SMA Sunny Island systems, thus facilitating rapid integration into operational environments [13,14]. However, these industrial demonstrators typically remain constrained by real-time testing protocols, which neither allow for accelerated simulation of typical operational days nor comprehensively cover diverse climatic scenarios and load profiles—both essential elements for representative validation in agricultural settings.

In parallel with physical validations, the optimized design of a microgrid requires a rigorous evaluation of its energy architecture. Numerous studies have compared AC and DC topologies based on various technical, economic, and operational criteria [15–18]. Nevertheless, these analyses typically focus on mixed-load scenarios, with little to no direct application to agricultural environments characterized exclusively by alternating current loads. This highlights a significant research gap, particularly for regions such as Morocco, where the specificity of energy use warrants differentiated evaluations. Furthermore, the literature increasingly emphasizes the optimal sizing of photovoltaic inverters, generally determined by a trade-off between investment costs (CAPEX) and overall energy efficiency [19–23]. However, existing models are predominantly developed under temperate (Ireland, Spain), tropical (Malaysia), Nordic (Finland), or continental (United States) climates, without explicitly integrating semi-arid climatic conditions and irradiation profiles specific to rural Moroccan regions.

These gaps identified in the literature, whether related to the transferability limitations of conventional experimental prototypes, the lack of comparative studies addressing AC architectures specifically in agricultural contexts, or the absence of sizing models tailored to Moroccan climatic conditions, underscore the necessity of developing an integrated methodology. Such a methodology must be experimentally

validated, adapted to regional specificities, and based on rigorous techno-economic considerations.

In this context, this study proposes an integrated and transferable framework for upgrading an isolated agricultural farm into a renewable microgrid under Moroccan rural constraints. The main contributions are as follows:

- A framework for upgrading fossil-fuel-based agricultural power systems to renewable microgrids in North Africa, structured as a complete workflow from site characterisation to design, sizing, and validation.
- A scalable AC microgrid prototype implemented with certified industrial-grade converters (Fronius, Victron Energy), ensuring transferability to real operational environments beyond conventional academic test benches.
- An accelerated-time experimental protocol that compresses a full operating day into a few hours while preserving realistic system dynamics using an actual agricultural load profile, enabling validation under representative daily scenarios rather than carefully selected short operating cases.
- Mathematical efficiency models developed to compare AC and DC coupling in the Moroccan context under the constraint of exclusively AC loads and an AC backup source, with comparative trends validated in MATLAB/Simulink, demonstrating the advantage of AC coupling for the considered case.
- A techno-economic optimisation of PV inverter sizing leveraging Moroccan climatic data to identify an appropriate PV-to-inverter ratio aligned with regional meteorological profiles.

The remainder of the paper is organized as follows. Section 2 describes the RIMSAR agricultural site and presents the full scale microgrid system design, including the load characterization and the sizing of the photovoltaic generator, the battery storage system, the coupling topology, and the associated power converters. Section 3 provides the economic and environmental assessment of the proposed solution. Section 4 details the design and experimental testing of a scaled microgrid prototype, covering the laboratory setup, the prototype sizing under accelerated test conditions, and the experimental results with their discussion. Finally, Section 5 summarizes the main conclusions and outlines perspectives for future developments.

2. Pilot farm and full-scale microgrid system design

This research is part of the Euro-African project “MG FARM - Smart stand-alone micro-grids as a solution for agriculture farms electrification”, co-funded by the European Union as part of the Horizon 2020 -

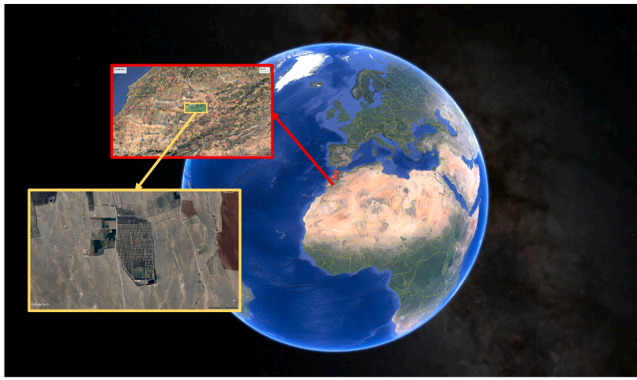


Fig. 2. Geographical Location of the RIMSAR Farm at Global and Regional Scales using Google Earth.

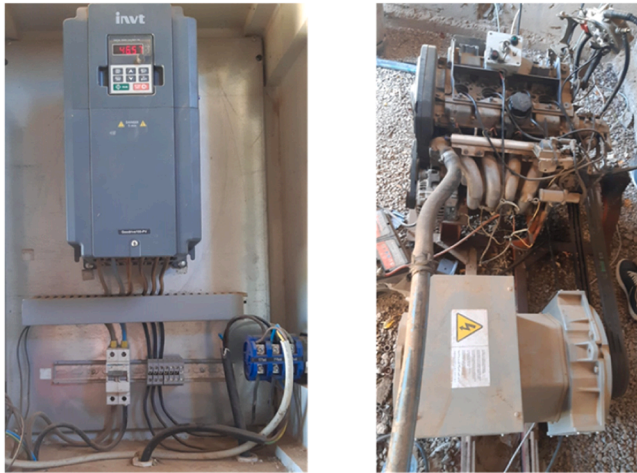


Fig. 3. Current Electrical Installation of the RIMSAR Farm Based on a Photovoltaic System and a Thermal Backup Source.

LEAP-RE program. The project aims to design and demonstrate innovative renewable energy-based solutions for the electrification and optimization of water management in the agricultural sector in North Africa, particularly in Morocco and Algeria [24].

In line with the objectives of the MG FARM project, the present study uses the RIMSAR farm, which is representative of rural agricultural operations in Morocco, as a pilot farm to develop a comprehensive methodology for energy conversion into an intelligent microgrid system.

2.1. Current configuration of the RIMSAR agricultural site

The actual installation of the RIMSAR farm is located in the rural municipality of Oulad Sbih, within the province of El Kelaâ des Sraghna, in the Marrakech-Safi region of Morocco. The figure below illustrates the geographical location of the site at both global and regional scales. [Fig. 2.](#)

As shown in [Fig. 3](#) and [Fig. 4](#), water pumping on the farm is currently provided by a direct photovoltaic system (PV-only system without storage), supplemented by a backup source consisting of a gasoline-powered thermal engine mechanically coupled to an AC synchronous generator.

The main load is associated with irrigation pumping, which operates intermittently depending on water demand and climatic conditions. Secondary loads, such as lighting or occasional agricultural equipment, may also be present but remain marginal.

This configuration represents a relevant case study for conversion to an intelligent microgrid. However, the current installation presents

several major drawbacks. It is notably characterized by the intensive use of a butane-powered thermal generator, which is a significant source of air pollution and prevents the optimal exploitation of available renewable resources. Furthermore, the absence of an Energy Management System (EMS) leads to substantial energy losses and low overall system efficiency. As a result, this configuration is not only environmentally detrimental but also generates particularly high capital and operational expenditures. These issues will be the subject of a detailed quantitative analysis in the following section dedicated to the system's economic and environmental assessment.

The objective is therefore to design a hybrid energy system integrating storage and centralized supervision, in order to optimize energy availability, reduce dependence on thermal generators, and maximize the use of renewable resources.

The next section presents the sizing process of the proposed microgrid, taking into account site-specific characteristics, load profile, available resources, and techno-economic constraints.

2.2. Electrical design of the farm microgrid system

This section presents the preliminary sizing of the various energy sources that will constitute the electrical installation of the farm microgrid. The system is scheduled to be deployed in January 2026 as part of the MG FARM project. The sizing is based on the analysis of the current electrical loads as well as the main agricultural activities observed on the site.

2.2.1. Load profile and agricultural activities of the farm

RIMSAR is a privately operated irrigated agricultural farm covering an area of approximately 12 hectares, with a perimeter of around 1 530 m. The main crops consist of olive trees of various varieties, along with a few citrus trees such as lemon and orange trees. [Fig. 5](#) illustrates some of the crops cultivated on the farm.

The agricultural farm is equipped with an irrigation system based on water pumping, powered by a photovoltaic installation that has been in operation since 2014. This system consists of a 17.2 kWp solar array made up of polycrystalline modules and does not include any battery storage. The solar energy produced is directly converted into electricity to supply a motor pump with a rated power of 15 kVA, as well as other low-power electrical loads such as lighting, environmental sensors, and energy meters.

Water is extracted from a well with a depth of 100 m and distributed through a drip irrigation system, which ensures localized water delivery at the root level. In parallel, the farm uses between three and four butane gas cylinders per day to operate a thermal engine that is mechanically coupled to a 30 kVA synchronous alternator, as shown in [Fig. 4](#).

[Fig. 6](#) presents the farm's electrical load profile over a one-year period, covering the interval from January 1st to December 31st, 2024. This profile makes it possible to characterize the overall energy consumption dynamics and its seasonal variations.

[Table 1](#) summarizes the main electrical loads identified on the farm, detailing their associated activities, rated power, estimated daily operating duration, and corresponding daily energy consumption. This information serves as a basis for the subsequent system sizing and energy performance assessment.

To further illustrate the intra-day variability of energy demand, the following figure presents typical daily consumption profiles representative of different seasons. [Fig. 7.](#)

The analysis of these daily profiles reveals significant seasonal variability, both in terms of usage time window and power demand. In winter, energy consumption is concentrated over a shorter period, whereas in summer and spring, the demand is more stable and extends throughout the day. These observations provide valuable insights for refining the energy source sizing strategies, which will be detailed in the following section.

Based on the identified consumption profiles, the following section

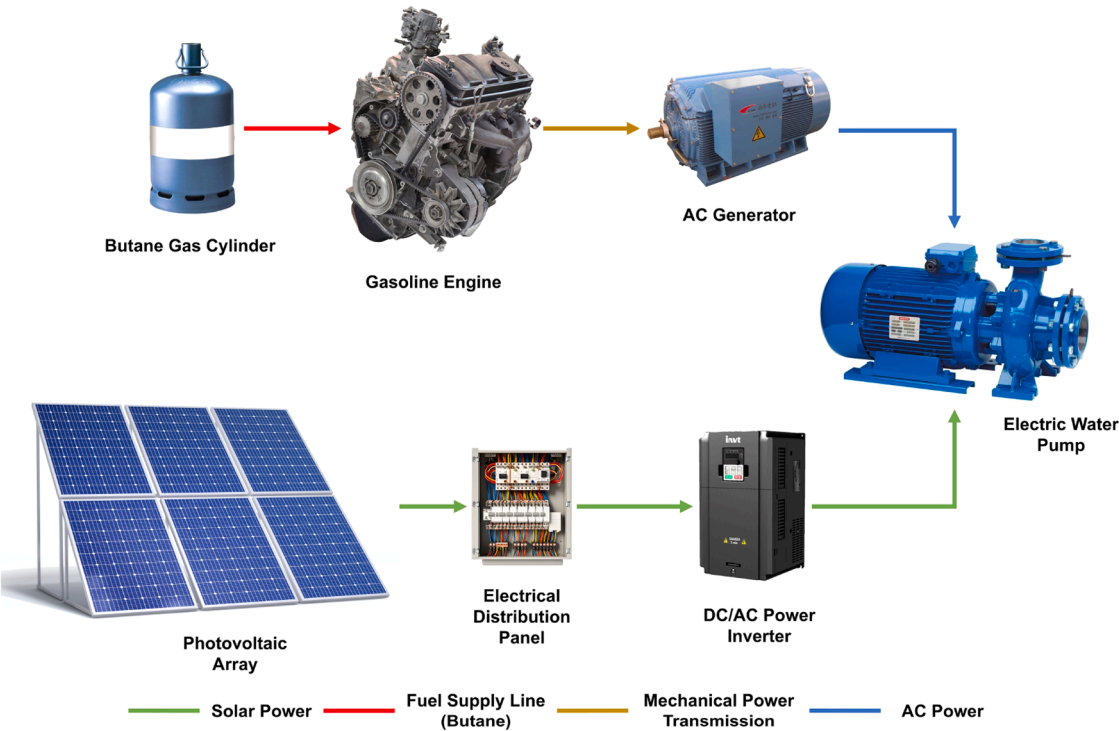


Fig. 4. Block Diagram of the Current Electrical Installation at the RIMSAR Farm.



Fig. 5. Crops Cultivated on the RIMSAR Farm: Olive Trees and Citrus Varieties.

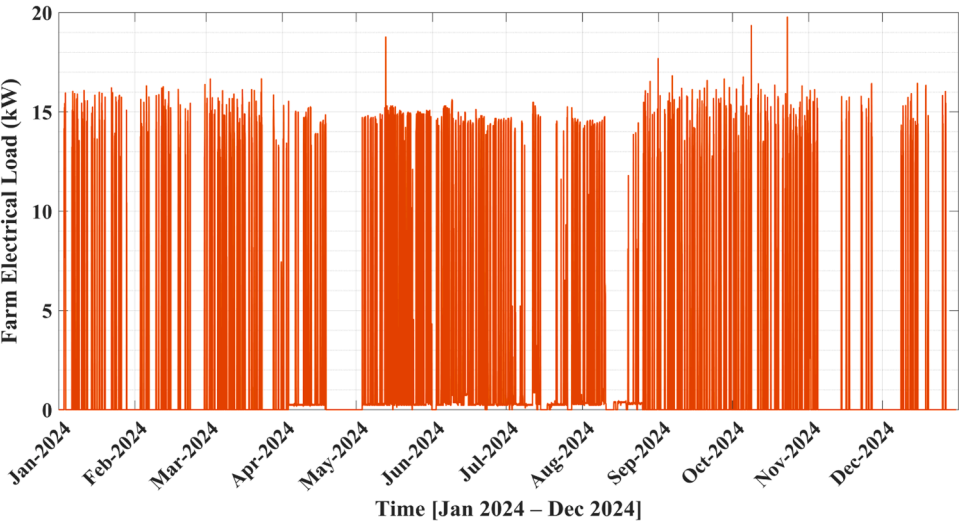


Fig. 6. Daily Electrical Load Profile of the RIMSAR Farm from January to December 2024.

Table 1

Main Electrical Loads of the RIMSAR Farm: Associated Activities, Power Ratings, and Daily Energy Consumption.

Equipment	Associated Activity	Average Peak Power (kW)	Average Daily Usage Duration (h/day)	Average Daily Energy Consumption (kWh/day)
AC Pump	Irrigation	15	9	135
Lighting + Sensors	Site Monitoring	0.15	24	3.6

focuses on the sizing of the photovoltaic system planned as part of the farm's transition toward a microgrid architecture.

2.2.2. The solar photovoltaic system

Photovoltaic solar energy is currently one of the main renewable sources deployed in agricultural energy systems, due to its local availability, modularity, and the significant decline in installation costs over the past decade [25–29]. However, its intermittent nature, linked to the variability of solar irradiation, limits its use in the absence of proper management or storage solutions [30–32].

In this study, a photovoltaic system is selected as the farm's primary energy source. This section describes the existing PV installation along with its technical characteristics and presents the sizing strategy adopted for the optimized system designed for the future conversion of the farm into an intelligent microgrid.

As shown in Fig. 8, and in accordance with the previously introduced elements, the existing installation—commissioned in 2014—consists of 66 photovoltaic modules. The technical specifications of a typical module are presented in Table 2.

In the context of converting the farm into an intelligent microgrid, a new sizing of the photovoltaic array will be carried out using specialized software. This sizing process is part of a broader strategy for optimizing the overall energy system and will be extended to the other sources and components of the microgrid in the following sections. The objective is to determine an optimized configuration that meets the farm's energy needs.

The sizing is based on the load profile representative of July 7 (Fig. 7), which is characterized by high and continuous demand typical of the summer season. Fig. 9 presents the results obtained using the PVSyst software, which is widely recognized for the analysis and sizing

of photovoltaic systems, particularly in the case of stand-alone configurations, which corresponds to the operating mode envisioned for the microgrid [33,34].

Based on the results, the currently installed PV capacity (17.2 kW) is 31.2 % below the PVSyst optimal value (25 kW). Combined with weather dependency and the absence of storage, this gap increases vulnerability and drives greater reliance on polluting butane or gasoline backup, which reduces energy autonomy and can negatively affect agricultural productivity. Accordingly, this study supports transitioning to an innovative microgrid architecture adapted to rural African farms to progressively reduce fossil fuel dependence and enhance system resilience and efficiency. The next section addresses storage sizing and justifies the selected technology within the proposed configuration.

2.2.3. Battery storage system

Energy storage is essential in standalone microgrids to ensure



Fig. 8. Current Photovoltaic Installation of the RIMSAR Farm.

Table 2

PV module's characteristics.

Trade mark, model, and type	SOLIMPEKS ENERG, SOL60P-260 W, polycrystalline module
Maximum Power at STC (P_{mpp})	260 W
Maximum Power Point Voltage (V_{mpp})	31.3 V
Maximum Power Current (I_{mpp})	8.31 A
Open Circuit Voltage (V_{oc})	38.2 V
Short Circuit Current (I_{sc})	8.99 A

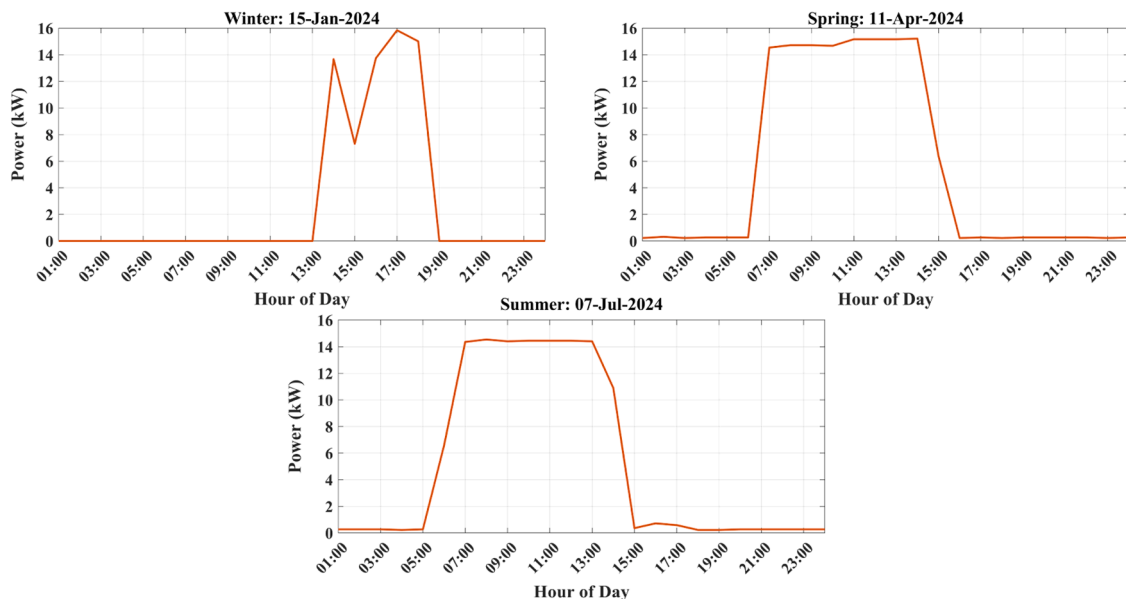


Fig. 7. Representative Daily Load Profiles for Different Seasons (Winter, Spring, Summer 2024).

PV module	
Manufacturer	Jinkosolar
Model	JKM410M-72H-V
(Original PVsyst database)	
Unit Nom. Power	410 Wp
Number of PV modules	60 units
Nominal (STC)	24.60 kWp
Modules	15 Strings x 4 In series
At operating cond. (50°C)	
Pmpp	22.51 kWp
U mpp	149 V
I mpp	151 A
Controller	
Universal controller	
Technology	MPPT converter
Temp coeff.	-5.0 mV/°C/Elem
Converter	
Maxi and EURO efficiencies	97.0 / 95.0 %
Total PV power	
Nominal (STC)	25 kWp
Total	60 modules
Module area	121 m ²
Cell area	107 m ²

Fig. 9. Photovoltaic System Sizing Results Using PVsyst for the RIMSAR Farm.

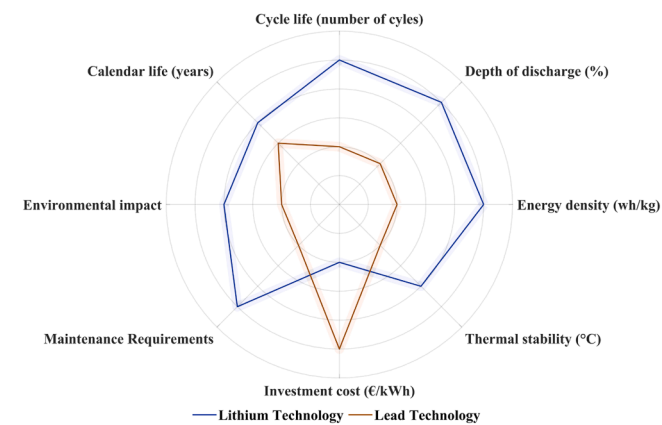


Fig. 10. Radar chart comparing lithium-ion and lead-acid batteries across eight technical and economic criteria relevant to rural microgrid design.

continuity of supply despite renewable intermittency [35,36]. In the proposed farm microgrid, the battery storage system is designed to stabilize electricity delivery during low irradiance periods and demand peaks [36–38]. Storage sizing is therefore defined to remain consistent with the daily load profile and the expected photovoltaic generation performance.

2.2.3.1. Comparative analysis of battery technologies for standalone microgrids. Storage technology selection critically influences performance, durability, and lifecycle economics in standalone microgrids. In practice, the most deployed families are lithium ion and lead acid batteries. In this study, Lithium Iron Phosphate (LiFePO₄) and gel type lead acid (GEL) are retained as the most appropriate options due to robustness, enhanced thermal stability, and operational reliability [39]. Lithium ion batteries provide higher energy density, enabling more

Table 3

Quantitative Comparison Between LiFePO₄ and Gel VRLA Battery Technologies (Victron Energy).

Criterion	LiFePO ₄ (LFP-12.8/200)	Gel VRLA (BAT412201104)
Energy Density (Wh/kg)	135	40
Depth of Discharge (%)	80	50
Cycle Life (cycles)	5,000	500
Investment Cost (€/kWh)	350	136
Thermal Stability (°C)	-20 to +70	Recommended: +20 to +40
Calendar Life (years)	> 12	12 (at 20 °C, floating mode)
Maintenance Requirements	Requires BMS and monitoring	No maintenance (sealed VRLA)

Battery

Manufacturer	Victron Energy
Model	LFP-CB 12.8V / 200Ah
Technology	Lithium-ion, LFP
Nb. of units	14 in parallel x 4 in series
Discharging min. SOC	10.0 %
Stored energy	127.2 kWh
Battery Pack Characteristics	
Voltage	51 V
Nominal Capacity	2828 Ah (C10)
Temperature	Fixed 20 °C

Battery Management control

Threshold commands as	SOC calculation
Charging	SOC = 0.96 / 0.80
Discharging	SOC = 0.10 / 0.35

Fig. 11. PVsyst Energy storage system sizing.

compact storage for constrained installations [39,40]. This characteristic is particularly advantageous for systems with space constraints or weight limitations. They also support deeper discharge ranges, which increases effective capacity utilization and reduces oversizing needs [38, 41,42]. In addition, lithium ion technologies offer longer cycle life than lead acid, and improved calendar life under proper operating conditions [38,39,41–43]. From an environmental and operational perspective, although lead acid recycling is widespread, improper handling carries risks linked to lead content, while lithium ion systems generally show a lower footprint but require dedicated recycling pathways for critical materials [39,42,44].

Lithium ion batteries also reduce maintenance requirements compared with lead acid systems [39–41]. Despite higher upfront costs, their longer lifetime and higher round trip efficiency can yield a more competitive leveled storage cost [38,41,42,45].

Finally, lead acid batteries are markedly temperature sensitive, with high temperatures accelerating corrosion and low temperatures reducing available capacity [39,40], whereas LiFePO₄ exhibits higher thermal stability and a wider operating range from -30 °C to +60 °C when suitable thermal management is implemented [39–41].

In summary, LiFePO₄ is retained as the preferred option for isolated microgrids due to its performance, lifetime, compactness, and thermal robustness, as summarized in Fig. 10: using a radar chart with scores normalized on a 0–5 scale based on the cited literature.

To complement this analysis with a concrete and context-specific evaluation, the following Table provides a quantitative comparison between two commercial battery models offered by Victron Energy. These models are representative of the Lithium Iron Phosphate

(LiFePO₄) and gel-type lead-acid (GEL) technologies previously discussed.

As shown in Table 3, the LiFePO₄ battery exhibits superior performance compared to the VRLA Gel battery across most technical criteria, particularly in terms of energy density, lifespan, and thermal stability. Based on this comparative analysis, the following section focuses on the sizing of the energy storage system.

2.2.3.2. Storage system sizing for the farm microgrid. Based on the findings of the literature review and the quantitative comparison of commercial models, Lithium Iron Phosphate (LiFePO₄) technology was selected as the energy storage solution for the farm's microgrid, due to its proven technical performance. Fig. 11 presents the sizing results obtained using the PVSyst software.

The selected architecture is based on 54 modules (14 in parallel $\times$ 4 in series) of the LFP-CB 12.8 V / 200 Ah model, yielding a total nominal capacity of 2,828 Ah at 51 V, equivalent to 127.2 kWh of stored energy.

The system is optimized for operation at a fixed temperature of 20 °C, with a minimum depth of discharge (DoD) set at 10 %, ensuring a balance between durability and performance. Charging and discharging processes are regulated by strict state-of-charge (SoC) thresholds (0.96/0.80 for charging and 0.10/0.35 for discharging), enabling safe and efficient battery utilization.

This configuration is designed to meet the farm's daily energy demand while ensuring long system lifespan and reliable energy autonomy.

After finalizing the sizing of the microgrid's sources, the focus shifts to the power converters interfacing sources and loads. Their selection is intrinsically linked to the coupling architecture, AC or DC, which determines the associated technical requirements and affects efficiency, complexity, and operational flexibility. The next section therefore compares both options to identify the most appropriate coupling approach for the farm under study.

2.2.4. Selection of coupling topology and sizing of power converters in the farm microgrid

Two coupling architectures are commonly considered for standalone microgrids, AC and DC. For the farm under study, both options are quantitatively compared using efficiency models that capture losses across the required power conversion stages, in order to justify the coupling architecture retained for converter sizing.

2.2.4.1. AC efficiency mathematical model for the AC configuration. In an AC-coupled microgrid, two converters are primarily involved: a grid-tied PV inverter interfacing the solar array with the AC bus, and a bidirectional inverter-charger enabling energy exchange between the battery and the AC network. Because the backup source considered here, the public utility grid, is inherently AC, no additional conversion stage is required when supplying the loads during local energy deficits. Accordingly, the system-level efficiency assessment accounts only for the PV and battery conversion stages.

The global efficiency model is given by Eq. (1), where $P_{pv,Abs}$ denotes the PV power prior to conversion, P_{loads} is the load demand, and P_{backup} is the backup supply during generation deficit. The parameters $\eta_{pv,inv}$ and $\eta_{discharge}$ represent the PV inverter efficiency and the discharge efficiency of the bidirectional converter, respectively.

$$\eta_{MG,AC} = \sum_{i=0}^4 \delta_{ij} \frac{P_{pv,Abs} \cdot \eta_{pv,inv} \cdot \Gamma_0 + (P_{loads} + (P_{PV,Abs} \cdot \eta_{pv,inv} - P_{loads}) \cdot \eta_i) \cdot (1 - \Gamma_0)}{P_{pv,Abs} + \delta_{i,3} \left(\frac{P_{load} - P_{pv} \cdot \eta_{pv,inv}}{\eta_{discharge}} \right) + \delta_{i,4} \frac{P_{backup} \cdot \Theta_{backup}}{\eta_{backup}}} \quad (1)$$

The complete set of parameters used in the model is defined in the system provided below.

$$\left\{ \begin{array}{l} j = \begin{cases} 0 & \text{if } \Delta P = 0 \\ \sum_{k=1}^4 k \Gamma_k & \text{else} \end{cases} \Rightarrow j = \begin{cases} 0 & \text{if } \Delta P = 0 \\ 1 & \text{if } \Delta P > 0 \wedge \text{SoC} \leq \text{SoC}_{\max} \\ 2 & \text{if } \Delta P > 0 \wedge \text{SoC} > \text{SoC}_{\max} \\ 3 & \text{if } \Delta P < 0 \wedge \text{SoC} \geq \text{SoC}_{\min} \\ 4 & \text{if } \Delta P < 0 \wedge \text{SoC} < \text{SoC}_{\min} \end{cases} \\ \Gamma_0 = \mathbf{1}_{\{\Delta P=0\}}; \Gamma_1 = H(\Delta P) \cdot H(\text{SoC}_{\max} - \text{SoC}) \cdot (1 - \mathbf{1}_{\{\Delta P=0\}}) \\ \Gamma_2 = H(\Delta P) \cdot (1 - H(\text{SoC}_{\max} - \text{SoC})) \cdot (1 - \mathbf{1}_{\{\Delta P=0\}}) \\ \Gamma_3 = H(-\Delta P) \cdot H(\text{SoC} - \text{SoC}_{\min}) \cdot (1 - \mathbf{1}_{\{\Delta P=0\}}) \\ \Gamma_4 = H(-\Delta P) \cdot (1 - H(\text{SoC} - \text{SoC}_{\min})) \cdot (1 - \mathbf{1}_{\{\Delta P=0\}}) \\ \Theta_{backup} = H(-\Delta P) \cdot H(\text{SoC}_{\min} - \text{SoC}) \cdot (1 - \mathbf{1}_{\{\Delta P=0\}}) \\ \eta_0 = 1 \text{ (idle)}; \eta_1 = \eta_{\text{charge}}; \eta_2 = \eta_{\text{injection}}; \\ \eta_3 = \eta_4 = 0 \text{ (unused)} \end{array} \right.$$

In this model, three logical functions are used to simplify the formulation of the operating conditions: the Heaviside function, denoted by $H(x)$, is defined as follows:

$$H(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

The Kronecker symbol, denoted by δ_{ij} :

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Finally, the indicator function $\mathbf{1}_{\{\Delta P=0\}}$, which equals 1 when $\Delta P = 0$, and 0 otherwise:

$$\mathbf{1}_{\{\Delta P=0\}} = \begin{cases} 1 & \text{if } \Delta P = 0 \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

The following section presents the equivalent model for a DC-coupled architecture, enabling a quantitative comparison with the previously developed AC-coupling approach.

2.2.4.2. Efficiency mathematical model for the DC configuration. In industrial practice, a DC coupled microgrid intended to supply AC loads typically relies on two key converters, namely an MPPT charge controller interfacing the PV source with the DC bus, and a DC to AC inverter connecting the DC bus to the AC bus. The inverter supplies the electrical loads and, when applicable, injects surplus power into the public grid. Accordingly, the overall efficiency of the DC coupled architecture is essentially determined by the performance of these two conversion stages. Eq. (5) provides the corresponding efficiency model while accounting for the operating scenario under consideration.

In this model, η_{mppt} , $\eta_{dc,ac}$ and $\eta_{discharge}$ denote the efficiency of the MPPT charge controller, the DC-AC inverter linking the DC and AC buses, and the discharge stage, respectively. However, since the considered DC architecture includes only two main conversion stages, the discharge efficiency is assumed to be unity $\eta_{discharge} = 1$. For consistency with the DC-bus charging path, the Mode 1 coefficient is defined as $\eta_1 = \eta_{\text{charge}} / \eta_{dc,ac}$. All other parameters used in the model are consistent with those previously defined for the AC coupling configuration.

$$\eta_{MG,DC} = \sum_{i=0}^4 \delta_{i,j} \left(\frac{P_{pv,Abs} \cdot \eta_{mppt} \cdot \eta_{dc,ac} \cdot \Gamma_0}{P_{pv,Abs} + \delta_{i,3} \left(\frac{P_{load} - P_{pv,Abs} \cdot \eta_{mppt} \cdot \eta_{dc,ac}}{\eta_{discharge} \cdot \eta_{dc,ac}} \right) + \delta_{i,4} \frac{P_{backup} \cdot \Theta_{backup}}{\eta_{backup}}} + \frac{(P_{loads} + (P_{pv,Abs} \cdot \eta_{mppt} \cdot \eta_{dc,ac} - P_{loads}) \cdot \eta_i) \cdot (1 - \Gamma_0)}{P_{pv,Abs} + \delta_{i,3} \left(\frac{P_{load} - P_{pv,Abs} \cdot \eta_{mppt} \cdot \eta_{dc,ac}}{\eta_{discharge} \cdot \eta_{dc,ac}} \right) + \delta_{i,4} \frac{P_{backup} \cdot \Theta_{backup}}{\eta_{backup}}} \right) \quad (5)$$

For readability, Table 4 summarizes the operating modes and the corresponding active conversion chains considered in Eqs. (1) and (5).

The two mathematical models developed to estimate the overall efficiency of the AC and DC coupling architectures are subsequently evaluated through a comparative simulation carried out in MATLAB/Simulink. The objective is to quantify the impact of the selected topological configurations on the energy efficiency of the microgrid system.

2.2.4.3. Comparative analysis and justification of the technology selection. The AC–DC comparison was performed in Matlab/Simulink using site-recorded 2024 meteorological data (solar irradiance and ambient temperature) and the daily load profile shown in Fig. 6. To capture seasonal variability, both architectures were evaluated over a full-year simulation. The following figure outlines the structure of the simulation model used for this analysis. Fig. 12.

Annual efficiencies account for topology-specific converter losses. The AC-coupled system uses a Fronius grid-connected PV inverter and a Victron Energy Quattro bidirectional inverter-charger, while the DC-coupled option uses a Victron Energy MPPT charge controller (peak efficiency up to 99 %) and a Victron MultiPlus-II DC/AC inverter. Matlab/Simulink simulations yield 95.25 % average annual efficiency for the AC architecture versus 94.03 % for the DC configuration, with monthly variations reported in Fig. 13.

As shown in Fig. 13, AC coupling outperforms DC coupling in all months, with a maximum efficiency gap of 2.1 % in April. Beyond efficiency, AC coupling offers greater flexibility and scalability for integrating additional generation, storage, or loads without major reconfiguration [45,46], and benefits from more standardized and widely available components that can reduce capital and maintenance costs in rural contexts [16,46,47]. Therefore, the AC-coupled architecture is retained for the farm microgrid, and the next section presents the sizing of the associated power converters.

2.2.5. Sizing of selected power converters for the farm microgrid architecture

The AC-coupled microgrid considered here includes two power conversion stages, a grid-connected PV inverter and a bidirectional inverter-charger interfacing the storage system with the AC bus. Prior

studies show that PV inverter sizing must balance CAPEX against the maximization of usable energy delivered to the grid, yet most approaches rely on generic climatic assumptions and rarely address regional conditions explicitly. To fill this gap for Moroccan operating conditions, this section introduces a region-specific techno-economic optimization framework that integrates local irradiation and temperature profiles to identify an optimal inverter sizing ratio for improved energy performance and economic viability. Accordingly, the inverter sizing ratio is defined as follows:

$$RS = \frac{P_{PV,STC}}{P_{inv,nom}} \quad (6)$$

Where $P_{PV,STC}$ denotes the rated PV array capacity under Standard Test Conditions (kWp), and $P_{inv,nom}$ present the nominal rated inverter capacity on the AC side (kVA). The following algorithm outlines the optimization procedure of this ratio based on a techno-economic framework:

Fig. 14.

The proposed algorithm explores a multi-objective design space via parametric Simulink simulations, considering energy yield, LCOE, and related techno-economic indicators across a range of inverter sizing ratios (RS). For each RS, the corresponding performance metrics are computed and the Pareto front is analysed to identify the “knee” point, which captures the most balanced compromise between maximizing energy yield and minimizing LCOE, since improvements beyond this point in one objective induce a disproportionate penalty in the other. Using a Pareto-distance criterion, the method thus selects a robust RS that jointly supports technical performance and economic viability. The following figure reports the techno-economic results by showing energy yield and LCOE versus RS, together with the reference points RS_{energy}, RS_{economic}, and RS_{knee} on the Pareto front. Fig. 15.

Fig. 15 identifies three reference inverter sizing ratios (RS): RS_{energy} = 0.9, RS_{economic} = 1.5, and RS_{knee} = 1.0. RS_{energy} maximizes the energy injected into the microgrid but implies higher CAPEX due to inverter oversizing, whereas RS_{economic} minimizes LCOE by reducing inverter capacity at the expense of a marked energy reduction driven mainly by clipping losses. RS_{knee} = 1.0 therefore provides the most balanced trade-off, maintaining satisfactory energy performance with moderate investment. Accordingly, for an installed PV capacity of 25 kWp, a 25 kW nominal PV inverter is recommended to support optimized operation under site-specific techno-economic conditions.

For the bidirectional inverter-charger, Victron Energy technical guidelines recommend a nominal rating at least equal to the PV inverter to ensure stable power exchanges between the AC bus, PV generator, and battery storage system [46–49]. The inverter-charger is therefore sized at 25 kVA, consistent with the PV inverter specification. The following Table summarizes the final sizing of the main sources and converters, including nominal ratings and key electrical characteristics based on commercially available product specifications.

As reported in Table 5, PV field sizing considered the existing on-site

Table 4
Operating modes and simplified expressions of $\eta_{MG,AC}(j)$ and $\eta_{MG,DC}(j)$ derived from Eqs. (1) and (5).

Mode	Operating condition	simplified expression of $\eta_{MG,AC}$ derived from Eq. (1)	simplified expression of $\eta_{MG,DC}$ derived from Eq. (5)
0	$\Delta P = 0$	$\eta_{MG,AC} = \eta_{pv,inv}$	$\eta_{MG,DC} = \eta_{mppt} \cdot \eta_{dc,ac}$
1	$\Delta P > 0 \wedge \text{SoC} \leq \text{SoC}_{\max}$	$\eta_{MG,AC} = \frac{P_{loads} + (P_{pv,Abs} \cdot \eta_{pv,inv} - P_{loads}) \cdot \eta_{charge}}{P_{pv,Abs}}$	$\eta_{MG,DC} = \frac{P_{loads} + \left(P_{pv,Abs} \cdot \eta_{mppt} - \frac{P_{loads}}{\eta_{dc,ac}} \right) \cdot \eta_{charge}}{P_{pv,Abs}}$
2	$\Delta P > 0 \wedge \text{SoC} > \text{SoC}_{\max}$	$\eta_{MG,AC} = \frac{P_{loads} + (P_{pv,Abs} \cdot \eta_{pv,inv} - P_{loads}) \cdot \eta_{injection}}{P_{pv,Abs}}$	$\eta_{MG,DC} = \frac{P_{loads} + (P_{pv,Abs} \cdot \eta_{mppt} \cdot \eta_{dc,ac} - P_{loads}) \cdot \eta_{injection}}{P_{pv,Abs}}$
3	$\Delta P < 0 \wedge \text{SoC} \geq \text{SoC}_{\min}$	$\eta_{MG,AC} = \frac{P_{loads}}{P_{pv,Abs} + \frac{P_{load} - P_{pv} \cdot \eta_{pv,inv}}{\eta_{discharge}}}$	$\eta_{MG,DC} = \frac{P_{loads}}{P_{pv,Abs} + \frac{P_{load} - P_{pv,Abs} \cdot \eta_{mppt} \cdot \eta_{dc,ac}}{\eta_{discharge} \cdot \eta_{dc,ac}}}$
4	$\Delta P < 0 \wedge \text{SoC} < \text{SoC}_{\min}$	$\eta_{MG,AC} = \frac{P_{loads}}{P_{pv,Abs} + \frac{P_{backup}}{\eta_{backup}}}$	$\eta_{MG,DC} = \frac{P_{loads}}{P_{pv,Abs} + \frac{P_{backup}}{\eta_{backup}}}$

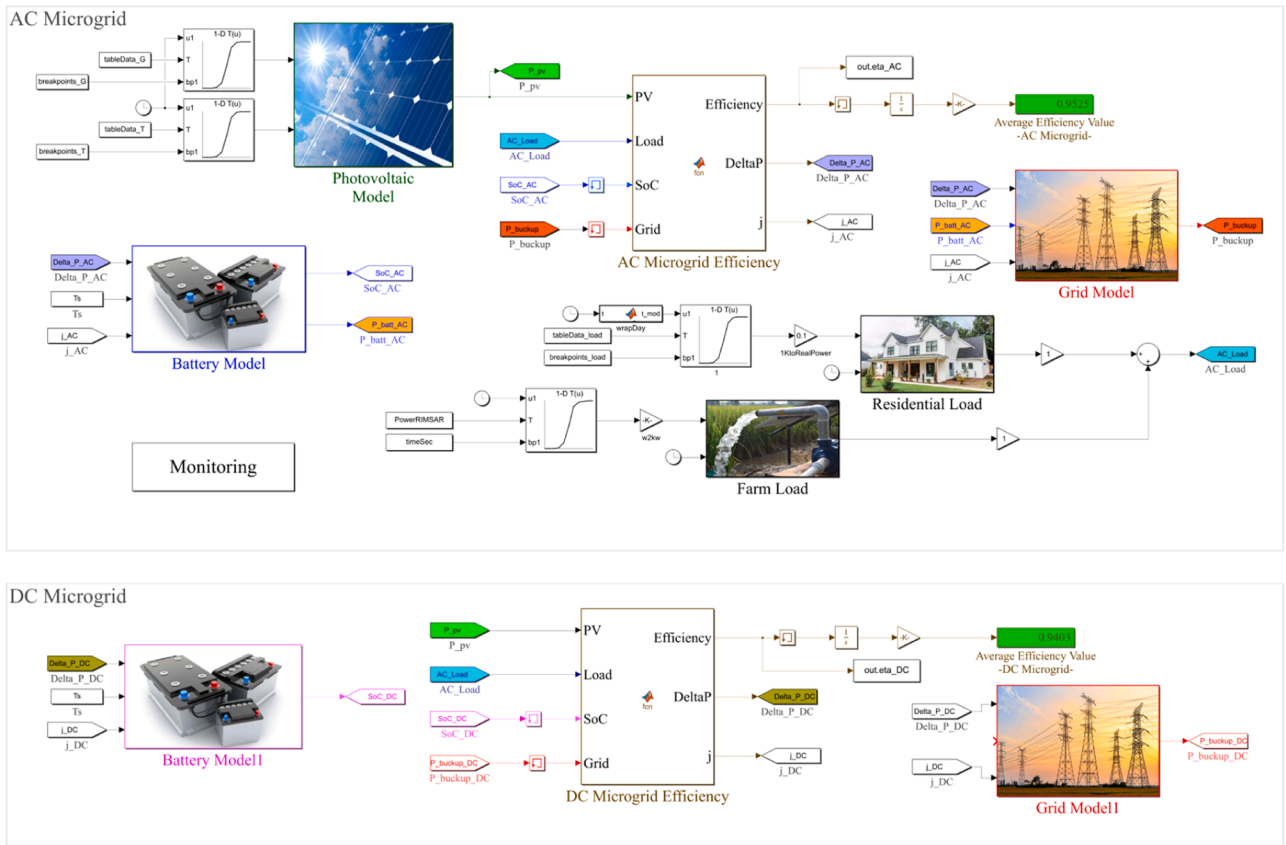


Fig. 12. Efficiency Modeling of AC and DC Microgrid Architectures in the MATLAB/Simulink Environment.

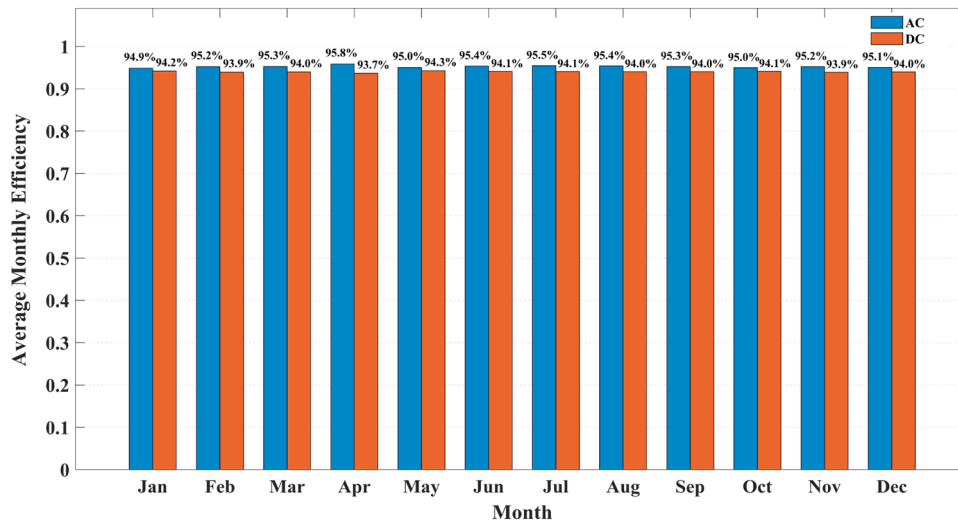


Fig. 13. Comparison of Annual Energy Efficiency Variation for AC and DC Microgrids.

generator, which was supplemented to reach the target capacity of 25 kWp. For the PV inverter, the design retained two separate units rather than a single 25 kW inverter for two practical reasons. First, the original PV array installed at the farm is no longer commercially available. Second, the Fronius ECO 25.0–3-S inverter, which could otherwise be considered, provides only one MPPT input and is therefore incompatible with the site requirement for two independent MPPT channels, one dedicated to the existing array and one to the new extension. Accordingly, using two inverters was identified as the most suitable solution. The next section presents the techno-economic and environmental

assessment to quantify the benefits of upgrading the current installation into a smart microgrid architecture.

3. Economic and environmental assessment

The primary objective of designing the microgrid system for the farm is to enhance the technical performance of the installation by ensuring a more stable, flexible, and efficient power supply, notably through the integration of an advanced energy management system. However, beyond these technical considerations, the project also aims to assess the

Algorithm: Techno-economic evaluation of PV-inverter sizing ratio (RS)

Input: Range of RS values, PV system parameters, inverter cost, weather data
Output: RS_energy, RS_economic, RS_knee

- 1 **Step 1: Initialization**
- 2 Define RS sweep range and constants (STC power, costs, lifetime);
- 3 **Step 2: Prepare simulation inputs**
- 4 **foreach** *RS in range* **do**
- 5 Configure Simulink model with current RS and weather data;
- 6 **Step 3: Run simulations in parallel**
- 7 Execute model for all RS values using Fast Restart;
- 8 **Step 4: Post-process results**
- 9 **foreach** *simulation result* **do**
- 10 Extract annual energy E_AC;
- 11 Compute yield $Y_f = E_{AC} / P_{STC}$;
- 12 Compute LCOE using CAPEX and E_AC;
- 13 **Step 5: Identify key RS values**
- 14 Find RS_energy where Y_f reaches a plateau;
- 15 Find RS_economic that minimizes LCOE;
- 16 Normalize Y_f and LCOE in $[0,1]$, invert Y_f ;
- 17 Find RS_knee using distance to Pareto front line;
- 18 **Step 6: Output RS values**
- 19 Return RS_energy, RS_economic, RS_knee;

Fig. 14. Pseudo-code of the techno-economic optimization of the PV-inverter sizing ratio (RS).

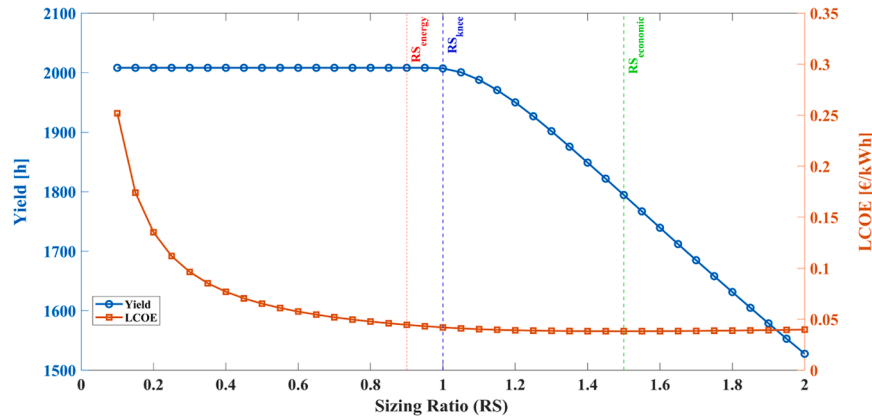


Fig. 15. Evolution of Energy Yield and LCOE as a Function of the Inverter Sizing Ratio (RS), with Identification of RS_energy, RS_economic, and RS_knee Operating Points.

Table 5

Summary of the selected components, brands, and key electrical characteristics for the designed agricultural microgrid.

Component	Brand	Characteristics
Photovoltaic system	SOLIMPEKS ENERGY, SOL60P-260 W, polycrystalline module Jinko Solar 72HC-550	66 modules $\times$ 260 W = 17.2 kWp 14 modules $\times$ 550 W = 7.7 kWp Total PV capacity $\approx$ 25 kWp
Energy Storage System	Victron Energy LFP- 12.8 V / 200Ah	21 modules in parallel $\times$ 4 in series = 84 modules 4242 Ah, 51 V
Photovoltaic Inverter	Fronius Symo Advanced 17.5–3-M Fronius Primo 8.2–1	17.5 kW + 8.2 kW $\rightarrow$ Total PV inverter capacity $\approx$ 25 kWp
Bidirectional inverter-Charger	Victron Energy Quattro	2 $\times$ 15 kVA = 30 kVA

potential economic and environmental benefits. This section is therefore dedicated to the joint analysis of these two complementary dimensions.

3.1. Economic analysis

This section compares the farm's existing energy system with the proposed optimized microgrid over a 25 year horizon by accounting for upfront investment, operation and maintenance (O&M) expenditures, and equipment replacements. The comparison is based on the Net

Present Cost (NPC), defined as the total discounted lifetime cost of the system and computed as follows.

$$\begin{aligned}
 \text{NPC} = & C_{\text{cap}} \\
 & + \sum_{m=1}^N \left[\frac{1}{(1+i_r)^m} \left(C_{\text{OM}} + \left(C_{\text{rep}} - C_{\text{rep}} \frac{\text{Rem}}{\text{Rep}} \right) \times \text{Integer} \left(\frac{n}{\text{Rep}} \right) \right) \right]
 \end{aligned} \quad (7)$$

where C_{cap} represents the initial investment cost, N is the project lifetime, and i_r is the real interest rate, with $i_r = \frac{i_n - f}{1 + f}$, where i_n is the nominal interest rate and f the inflation rate. C_{OM} denotes annual O&M cost, C_{rep} is the replacement cost, Rem is the remaining lifetime at the end of the project, and Rep is the component technical lifetime. The project duration is n , typically equal to N , and $\text{Integer}\left(\frac{n}{Rep}\right)$ gives the number of full replacements over the horizon. Input assumptions for the existing installation and the proposed microgrid are summarized in Table 6 and Table 7, respectively.

The cost evaluation is performed at the overall system level using empirical data and actual consumption records, given the absence of reliable unit cost references expressed in €/kW. O&M costs are estimated over the full lifetime of each component. For the backup source, the annual energy exchange cost with the public grid is computed as follows.

$$C_{grid,energy} = \sum_{i=1}^{N_{tarifs}} \sum_{j=1}^{12} E_{gp,i,j} \cdot C_{buy,i} - \sum_{i=1}^{N_{tarifs}} \sum_{j=1}^{12} E_{gs,i,j} \cdot C_{sell,i} \quad (8)$$

In Eq. (8), $E_{gp,i,j}$ is the purchased energy and $C_{buy,i}$ is the unit price per tariff tier defined by ONEE [50]. The second term represents potential revenue from injected energy $E_{gs,i,j}$ at the corresponding feed in tariff $C_{sell,i}$.

Under Law No 82 21 on self-generation of electricity (Official Bulletin No 7400, May 1st, 2025) [51], prosumers may inject up to 20 % of annual production, but financial compensation remains conditional on the publication of a feed in tariff by ANRE. Since this tariff and the associated technical regulations were not yet in effect at the time of the study, the RIMSAR microgrid is operated under a zero injection policy, which removes the second term of Equation (8) and yields.

$$C_{grid,energy} = \sum_{i=1}^{N_{tarifs}} \sum_{j=1}^{12} E_{gp,i,j} \cdot C_{buy,i} \quad (9)$$

Regarding the contribution of the backup source to meeting the farm's load during outage events, the following figure illustrates the monthly variation in backup energy over a full year. These results are derived from the simulation presented in Fig. 12, using a real load profile shown in Fig. 6, region-specific meteorological data, and the detailed component sizing described in the preceding sections.

The results presented in Fig. 16 were subsequently used to calculate the total annual cost of backup energy, using a MATLAB script. The underlying computation logic is summarized by the pseudocode illustrated in Fig. 17.

The calculation performed based on the pseudocode yields a value of 1,335.09 €, which corresponds to the annual consumption cost of the backup source over 25 years, as reported in Table 7 : Economic

parameters and lifetime assumptions of the proposed PV-battery-based microgrid. Based on the previously summarized technical and economic assumptions, the following figure illustrates a comparison of the Net Present Cost (NPC) between the conventional installation and the proposed microgrid architecture.

Fig. 18 compares the resulting NPC over 25 years. The conventional option requires a low initial investment of 9.6 k€ but reaches approximately 130.3 k€ due to recurring fuel and maintenance costs. The PV battery microgrid involves higher upfront CAPEX of about 62.1 k€ but stabilizes at 104.6 k€, yielding a reduction of 25.7 k€ over the project lifetime.

3.2. Environmental analysis

In addition to the economic assessment, an environmental analysis is conducted to quantify the impact of the proposed energy transition on greenhouse gas emissions. The current installation partially relies on a thermal engine powered by butane gas, whose combustion significantly contributes to CO₂ emissions.

This section aims to estimate the current emissions associated with the annual butane consumption and to compare these results with the expected environmental performance of the optimized microgrid system, which is primarily powered by a photovoltaic source. The objective is to evaluate the potential carbon footprint reduction over a 25-year time horizon. The total greenhouse gas emissions are evaluated using the following equation:

$$GHG_{total} = \sum_{i=1}^n GHG_i \quad (10)$$

Where GHG_{total} is the total greenhouse gas emissions (in kg CO₂-eq), GHG_i is the amount of emissions generated by the i th energy source (in kg CO₂-eq), n is the total number of emission sources considered and i is the index of each source.

In the case of the farm under consideration, the energy system relies on two sources: a photovoltaic source and a thermal source powered by butane gas. The distribution of the supplied energy is expressed by the following equation:

$$E_{loads}(t) = E_{PV}(t) + E_{gasoline_{motor}}(t) \quad (11)$$

Since only the thermal source is responsible for direct greenhouse gas emissions, the environmental analysis can be limited to this component, thereby allowing the emissions assessment to be simplified as follows:

$$GHG_{total} = GHG_{LPG} = Fuel_{consumption_{LPG}} \times EF_{CO_2, LPG} \quad (12)$$

With GHG_{LPG} is the greenhouse gas emissions due solely to butane fuel combustion (in kg CO₂-eq), $Fuel_{consumption_{LPG}}$ is the total quantity of

Table 6

Economic parameters and lifetime assumptions of the conventional agricultural pumping installation.

Component	Capital cost	O&M cost	Replacement cost	Lifetime
Gasoline engine	700 €	750 €	700 €	5 years
Butane fuel	110 €	159,688 €	-	25 years
AC generator	3,500 €	0 €	3,500 €	15 years
PV modules	4,620 €	1,500 €	0 €	25 years
PV inverter	660 €	125 €	660 €	15 years

Table 7

Economic parameters and lifetime assumptions of the proposed PV-battery-based microgrid.

Component	Capital cost	O&M cost	Replacement cost	Lifetime
PV modules	1,680 €	1,500 €	0 €	25 years
LiFePO ₄ battery	49,280 €	0 €	49,280	13 years
PV inverter	2,775 € + 1,900 €	125 €	2,775 € + 1,900 €	15 years
Battery inverter/charger	3,226,90 × 2	125 €	3,226,90 × 2	15 years
Backup	-	1,335 €	-	25 years

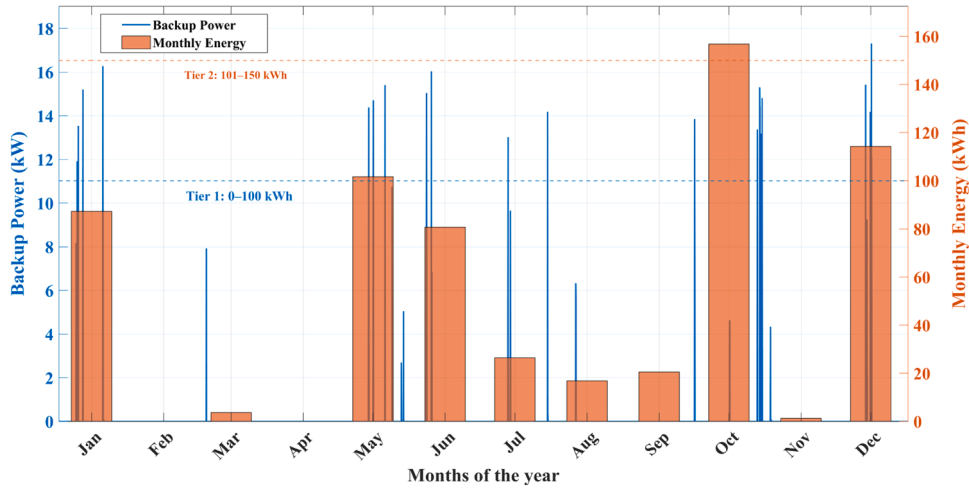


Fig. 16. Monthly Variation of Backup Power and Energy Consumption Over One Year.

Algorithm: Annual Backup Energy Cost Calculation (TTC)

Input: $P_{\text{backup}}(t_k)$: backup power over time (kW),
tariff_limits[1:6]: tranche boundaries (kWh),
price_TTC[1:6]: TTC price per tranche (€/kWh)
Output: $C_{\text{grid,energy}}$: total annual cost of backup energy (TTC, €)

```

1  $C_{\text{grid,energy}} \leftarrow 0$  // initialize total cost
2 for  $m \leftarrow 1$  to 12 do
3    $E_m \leftarrow \text{mean}(P_{\text{backup}})_m \times \text{hours\_in\_month}(m)$  // energy in kWh for month m
4   /* Allocate  $E_m$  into each of the 6 tariff tranches */
5    $E_1 \leftarrow \min\{E_m, \text{tariff\_limits}(1)\}$ ;
6    $E_2 \leftarrow \min\{\max(E_m - \text{tariff\_limits}(1), 0), \text{tariff\_limits}(2) - \text{tariff\_limits}(1)\}$ ;
7    $E_3 \leftarrow \min\{\max(E_m - \text{tariff\_limits}(2), 0), \text{tariff\_limits}(3) - \text{tariff\_limits}(2)\}$ ;
8    $E_4 \leftarrow \min\{\max(E_m - \text{tariff\_limits}(3), 0), \text{tariff\_limits}(4) - \text{tariff\_limits}(3)\}$ ;
9    $E_5 \leftarrow \min\{\max(E_m - \text{tariff\_limits}(4), 0), \text{tariff\_limits}(5) - \text{tariff\_limits}(4)\}$ ;
10   $E_6 \leftarrow \max\{E_m - \text{tariff\_limits}(5), 0\}$ ;
11  /* Compute monthly cost at TTC rates */
12   $\text{cost}_m \leftarrow \sum_{i=1}^6 E_i \times \text{price\_TTC}(i)$ ;
13   $C_{\text{grid,energy}} \leftarrow C_{\text{grid,energy}} + \text{cost}_m$ ;
14 return  $C_{\text{grid,energy}}$ 

```

Fig. 17. Pseudo-code of the annual backup energy cost calculation (TCC).

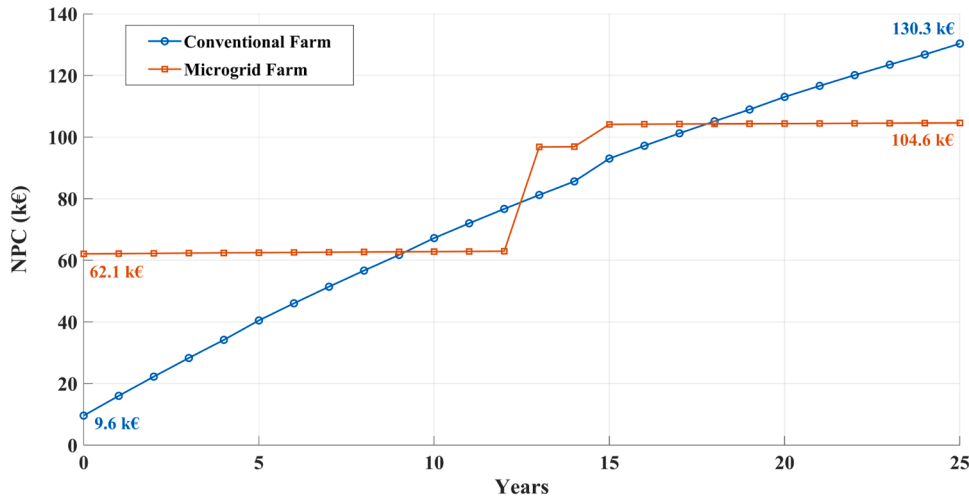


Fig. 18. Comparison of the Net Present Cost (NPC) between the Conventional Farm and the Proposed Microgrid Farm.

butane consumed during the system operation (in TJ), and $EF_{\text{CO}_2, \text{LPG}}$ is the CO_2 emission factor of butane (in $\text{kg CO}_2/\text{TJ}$), as defined by the IPCC guidelines.

In the farm under study, an average of 3.5 gas cylinders is used daily to meet the energy demand. Each cylinder contains 12 kg of LPG, resulting in an estimated annual consumption of 15,330 kg/year. To

apply Equation (12), this mass must be converted into energy (in TJ) using the fuel's energy content, as defined by the following equation:

$$\text{Fuel}_{\text{consumption}_{\text{LPG}}}(\text{TJ}) = \text{Mass}(\text{kg}) \times \text{NCV}(\text{TJ} / \text{kg}) \quad (13)$$

In this equation, $\text{Fuel}_{\text{consumption}_{\text{LPG}}}(\text{TJ})$ represents the total energy content of the fuel, $\text{Mass}(\text{kg})$ is the amount of fuel consumed, and NCV

(TJ/kg) is the net calorific value of the fuel, as defined by the IPCC guidelines.

According to the 2006 IPCC Guidelines (Vol. 2, Table 1.2) [52], the Net Calorific Value (NCV) of LPG is 47.3 MJ/kg (0.0000473 TJ/kg). Therefore, corresponds to an energy content of 0.7251TJ/years.

By applying Eq. (12) and the IPCC recommended CO₂ emission factor for LPG combustion (Vol. 2, Table 2.5), $EF_{CO_2, LPG} = 63,100 \text{ kg CO}_2 / \text{TJ}$, the annual emissions associated with the conventional installation are estimated at 45,764 kg CO₂/year. This value is used as the baseline for the environmental comparison with the proposed microgrid architecture for the RIMSAR farm in El Kelâa des Sraghna, in which the national grid is only used as a backup source.

In addition, based on the official report of the Moroccan National Office of Electricity and Drinking Water [53], the average grid emission factor is 744.6 gCO₂/kWh (i.e., 0.745 kgCO₂/kWh). Considering the previously reported average backup power of 81.29 W, the associated annual grid related emissions are estimated at 530.51 kg CO₂/year.

Based on these results, the following figure provides a graphical comparison between the two architectures. Fig. 19.

The figure evidences the substantial environmental gains achieved by replacing the conventional butane-based thermal system with the proposed microgrid architecture, supported by limited public-grid backup. Under the current configuration, emissions are approximately 45.76 tCO₂/year, corresponding to an estimated 1,144.1 tCO₂ over 25 years. In contrast, the PV-battery microgrid reduces emissions to about 0.53 t/year, totaling 13.26 tCO₂ over the same horizon, which corresponds to a greenhouse-gas reduction exceeding 98 %. These results highlight the strong environmental value of renewable microgrids for agricultural operations. Building on these economic and environmental outcomes, a downscaled laboratory prototype was developed to experimentally validate operational feasibility under realistic constraints. The next sections present its theoretical sizing and the experimental results obtained under accelerated testing conditions.

4. Design and experimental testing of a scaled microgrid prototype

To validate the different operating scenarios as well as the effectiveness of the control strategy developed, a microgrid prototype was designed and implemented in the laboratory as part of the MG-FARM project.

4.1. Description of the laboratory-scale prototyping system

This prototype is a functional replica of the microgrid architecture envisioned for the farm, but scaled down in terms of power. It also offers the advantage of operating under accelerated time conditions, enabling the simulation of an entire day of operation within just a few hours, as will be detailed in the following sections.

This part of the article focuses on presenting the equipment implemented in the prototype, prior to discussing the specific sizing strategy adopted for the experimental campaign. The following figure shows the schematic architecture of the laboratory prototype to illustrate its overall configuration. Fig. 20.

In this prototype, the photovoltaic source is represented by a photovoltaic emulator capable of reproducing the behavior of a real PV panel, such as the one already installed on the farm, or the Jinko Solar 72HC-550 modules planned for system expansion. This approach ensures independence from weather conditions and allows the use of the same irradiance and temperature data recorded on the agricultural site, thereby enabling test conditions that accurately reflect real-world operation.

Regarding the energy storage system, the choice of technology is assessed in light of the objective of the experimental campaign. The experiments primarily aim to validate the operational feasibility of the proposed architecture and the associated energy management logic

under accelerated-time conditions, with a specific emphasis on battery energy availability to supply the load, ensure supply continuity, and preserve the consistency of operating modes within a constrained experimental protocol. In this context, the electrochemical advantages of LiFePO₄ that motivate its recommendation for the full-scale installation mainly pertain to long-term performance attributes (service life, cycle life, energy density, and thermal sensitivity) and are not decisive with respect to the above indicators evaluated on the prototype.

Accordingly, considering the accelerated testing constraints and local availability, lead-acid storage technology was selected for the prototype.

The converters used are of the same brands listed in Table 5, but in reduced-power versions, consistent with the prototype sizing discussed later. This choice ensures the transferability of the experimental results to the real system by maintaining identical communication protocols and control logic between the converters and the microgrid controller. In addition, the Victron Energy VE.Bus platform supports parallel and three phase configurations, allowing the bidirectional inverter charger stage to scale up to 180 kVA in a three-phase system using 15 kVA Quattro units, with up to four units per phase. On the PV side, scalability can be achieved by connecting multiple Fronius inverters in parallel on the AC bus, each one of them could be rated at 125 kW. For higher power levels, scalability can be further addressed through a hierarchical approach in which multiple VE.Bus subsystems are deployed, each interfaced with its own controller, and coordinated at the supervisory EMS level. More generally, the proposed workflow of the section 2 is data driven and can be transferred to other African settings by updating the site-specific load profile and the local meteorological time series, which in turn updates the sizing of energy sources and power converters while preserving the same control architecture and evaluation procedure.

For the backup source, a grid simulator was implemented to emulate the behavior of the national electrical grid and to introduce disturbances such as voltage sags, frequency deviations, or phase imbalances, thereby contributing to the robustness testing of the microgrid.

Finally, overall system control is managed by a Raspberry Pi, which communicates with the various meters via the RS485 protocol and controls the converters in real time to manage energy flows within the system.

Following this general overview of the test bench components and architecture, the next section details the theoretical sizing adopted for the prototype, in line with the objectives of experimental validation.

4.2. Laboratory prototype sizing

For the laboratory-scale sizing, the prototype is conceived as a functional replica of the agricultural farm, scaled down in power. The load profile used for sizing is derived from the real farm demand and reduced by a factor of ten to comply with the test bench constraints. The following equation is then used to determine the peak power of the photovoltaic generator.

In this expression, $P_{load}(t_i)$ denotes the load demand at time t_i (in kW), while $G(t_i)$ et $T_c(t_i)$ represent the irradiance (in W/m²) and cell temperature (in °C), respectively, at the same time instant. The terms G_{STC} and T_{STC} refer to the standard test conditions, set at 1,000 W/m² and 25 °C. Finally, α_p is the temperature coefficient of the maximum power, and η_{sys} is the overall system efficiency.

$$P_{PV} = \frac{\sum_{t_i=1}^{24} P_{load}(t_i)}{\eta_{sys} \cdot \sum_{t_i=1}^{24} \frac{G(t_i)}{G_{STC}} [1 + \alpha_p (T_c(t_i) - T_{STC})]} \quad (14)$$

Applying the sizing equation results in a peak PV power of 2.06 kW for the laboratory-scale prototype, associated with a simulated daily energy demand of 1.54 kWh compressed into a three-hour test window through temporal acceleration, where 1 h of real operation corresponds to 7.5 min of test time. This accelerated-time validation strategy,

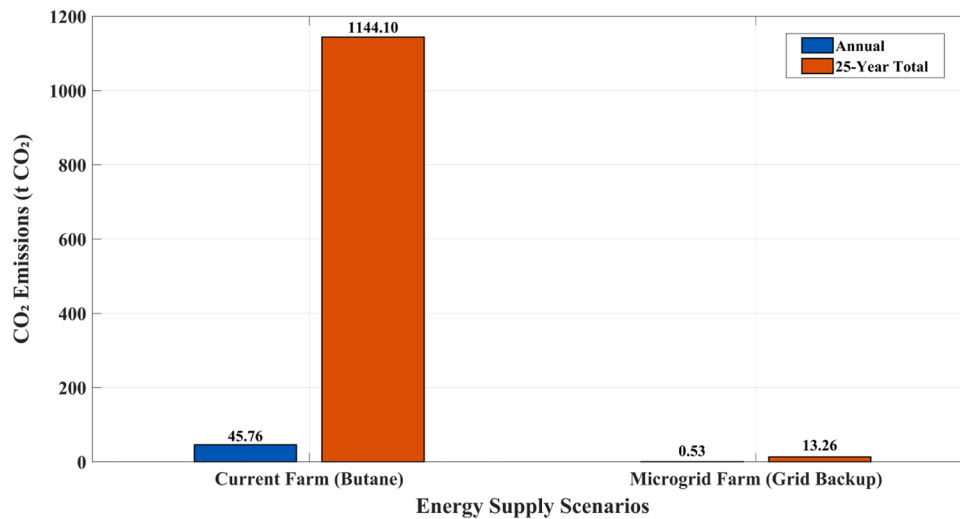


Fig. 19. Annual and cumulative 25-year CO₂ emissions comparison between the current butane-based farm and the proposed microgrid architecture with grid backup.

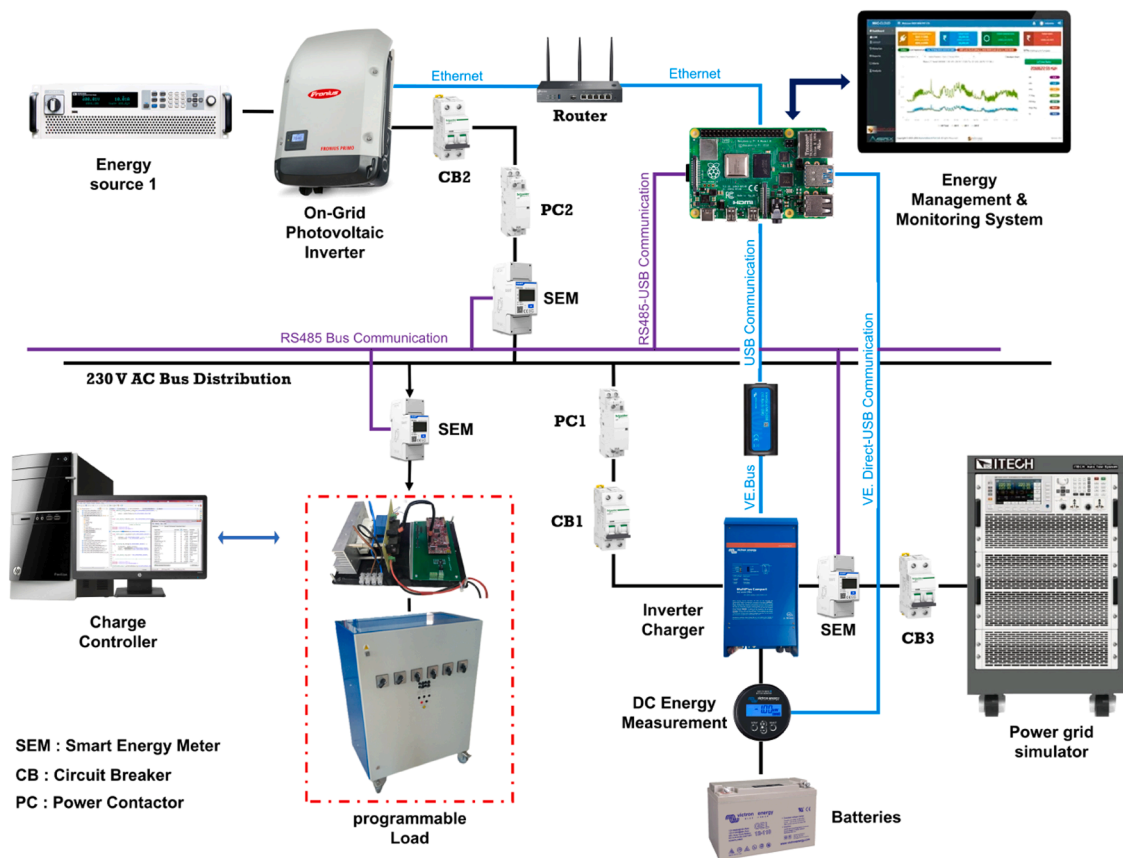


Fig. 20. Electrical diagram of the Laboratory-Scale Prototype.

embedded directly within the sizing stage, constitutes a key methodological asset by enabling rigorous performance assessment while reducing experimental duration and resource requirements. In parallel, the energy storage system is sized to ensure supply continuity and to faithfully reproduce daily load scenarios under the same accelerated conditions, as detailed in the following subsection.

$$Q_T = \frac{\sum_{i=1}^{24} P_{\text{load}}(t_i) \cdot \frac{T_{\text{twin}}}{24}}{N_S \cdot V_{\text{bat, nom}} \left(1 - \frac{\text{SoC}_{\text{min}}}{100} \right)} \quad (15)$$

With, $P_{\text{load}}(t_i)$ representing the load power at time step i , T_{win} denoting the reduced time window of the accelerated laboratory test, and SoC_{min} indicating the minimum state of charge. Furthermore, $V_{\text{bat, nom}}$ is the nominal voltage of a battery cell, and N_s is the number of cells connected in series to form the total pack voltage, set here to 24 V. The numerical application of this equation leads to a total rated capacity of 120 Ah for the prototype, corresponding to a usable energy of 3.85 kWh at a nominal voltage of 24 V.

Finally, for the power converters, the same sizing principles defined for the full-scale farm remain applicable to the prototype. The following

Table summarizes the characteristics of the laboratory-scale prototype, which is intended for the experimental validation of the proposed microgrid architecture for the farm. [Table 8](#).

Following the sizing of the laboratory prototype, the next section presents the experimental results obtained, along with a detailed discussion.

4.3. Experimental results and discussion

After describing and sizing both the real farm and the downscaled laboratory prototype operating under accelerated time conditions, this section presents the experimental validation of the microgrid prototype developed for the RIMSAR site. To this end, the following figure illustrates the test bench used, including the various components previously detailed in the preceding section. [Fig. 21 and 22](#).

Prior to detailing the experimental results, [Fig. 23](#) presents a flowchart illustrating the energy management strategy implemented within the laboratory microgrid. This diagram differentiates the various operating modes according to (i) renewable source availability, (ii) the storage system's state-of-charge (SoC), and (iii) the instantaneous load requirements. In [Fig. 23](#), black dots denote connection nodes used to indicate flow continuation between separated blocks and to avoid long or crossing arrows.

In this flowchart, P_{diff} denotes the difference between the available renewable power (P_{RES}) and the power demanded by the loads (P_{loads}). The thresholds SoC_{min} et SoC_{max} respectively represent the minimum and maximum allowable state-of-charge limits for the battery, while P_{Backup} indicates the power supplied by the backup source. Finally, P_{diff2} corresponds to the residual surplus after accounting for the allowable power intended for battery charging; for lead-acid batteries, this charging power is constrained by technical limits aimed at preserving

battery longevity, expressed by the following relationship (see [Eq. \(16\)](#)).

$$P_{Bat, Charge} = 0.2C_{Bat} \times V_{Bat, Bulk} \quad (16)$$

Where C_{Bat} denotes the capacity of the battery bank, and $V_{Bat, Bulk}$ represents the voltage during the bulk charging phase.

Based on the energy management strategy previously described, several experimental tests were conducted to assess the dynamic performance of the microgrid across various representative scenarios. [Fig. 24](#) illustrates the results of the first scenario, corresponding to a sunny day with the storage system fully charged at the outset. In this case, the applied load profile replicates the typical consumption pattern of an irrigation pump during the month of July, as presented in [Fig. 7](#).

The analysis of experimental results confirms the proper execution of the operational sequences outlined by the energy management strategy, demonstrating effective coordination between various energy sources based on photovoltaic availability and the state-of-charge (SoC) of the storage system. Load activation at 37.5 min (equivalent to 5:00 AM in real-time) is initially ensured by the battery due to the absence of solar generation. Once photovoltaic power becomes sufficient and surpasses the demand with a peak recorded around 400 W within the interval [00:13–01:53] the excess energy is directed toward recharging the battery, in compliance with safety constraints associated with lead-acid battery technology (see [Eq. \(16\)](#)), provided the SoC remains below its maximum threshold. After reaching the maximum allowable charging power, as observed between 01:53 and 02:12, surplus energy is redirected to the grid or injected into a neighboring microgrid in the farm's actual configuration.

Regarding the second scenario illustrated in [Fig. 25](#), the storage system initially exhibits a low state-of-charge, starting at 62.5 %. Load activation at 37.5 min is thus provided by the battery, resulting in a discharge current of approximately 0.7 C, given the absence of

Table 8

Laboratory-scale prototype components for experimental validation of the proposed microgrid architecture.

Component	Brand	Characteristics
Photovoltaic system	Itech Electronics PV Emulator	2 kWp
Energy Storage System	Victron Energy Gel	160,4 Ah - 24 V - 3,85 kWh
Photovoltaic Inverter	FRONIUS GALVO 2.0-1	2 kW
Bidirectional Inverter-Charger	Victron Energy MultiPlus	2 kW



Fig. 21. Laboratory-Scale Prototype: (1) Photovoltaic Emulator & Interface, (2) Grid Simulator, (3) Fronius On-Grid PV Inverter, (4) Energy Management System Controller, (5) AC Loads, (6) Battery Monitor, (7) Communication Router, (8) Hioki Power Quality Analyzer.



Fig. 22. Electrical Cabinet: (1) MultiPlus Inverter-Charger, (2) Gel Battery Bank, (3) AC Smart Meters, (4) Smart Shunt, (5) Electrical Protection Devices.

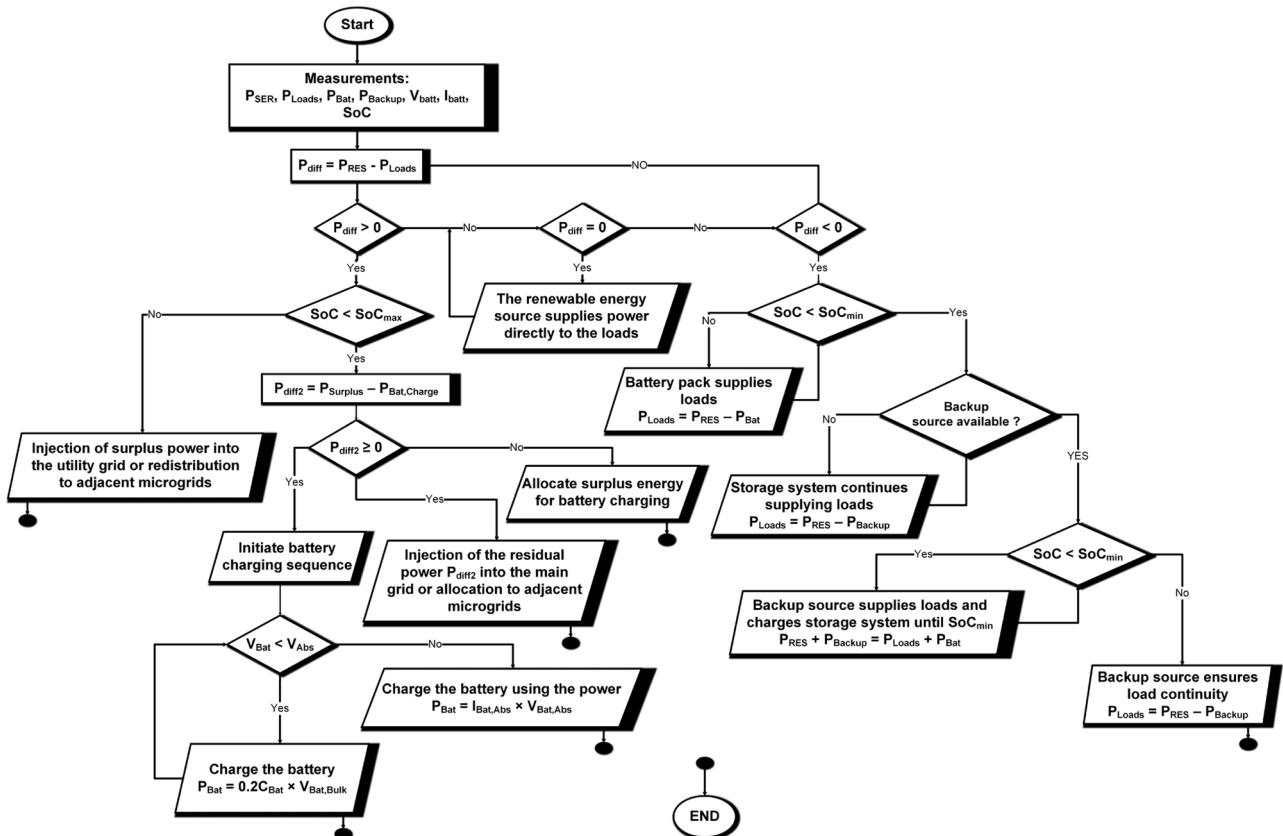


Fig. 23. Energy Management Strategy Implemented in the Laboratory Microgrid Prototype.

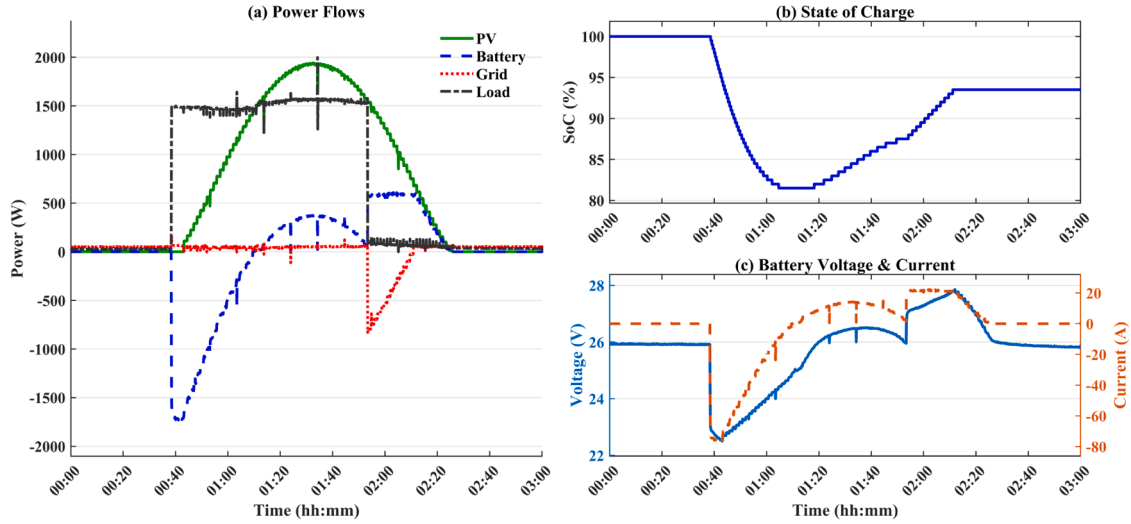


Fig. 24. Experimental Results – Scenario 1: Sunny Day with Fully Charged Battery.

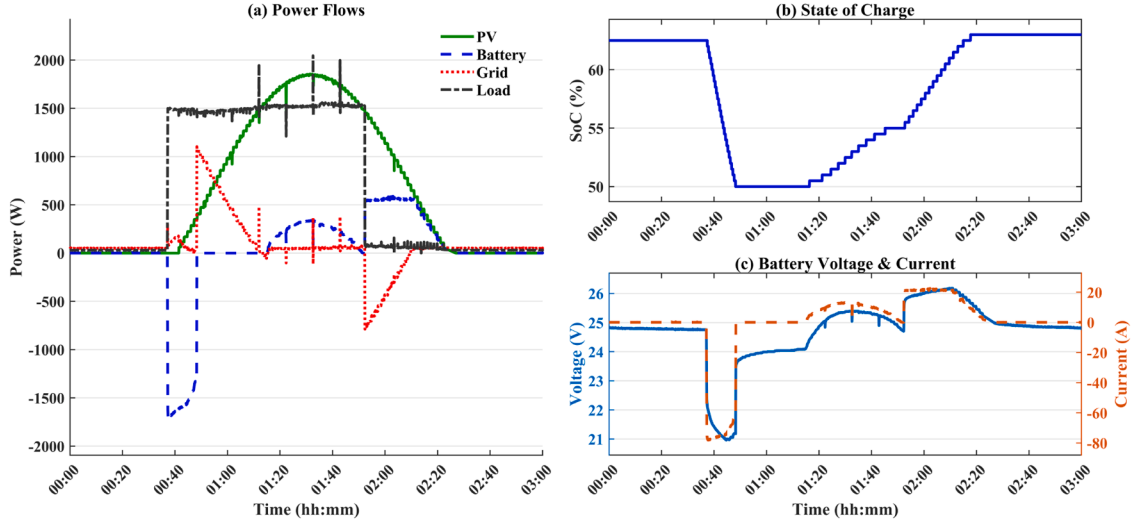


Fig. 25. Experimental Results – Scenario 2: Sunny Day with Initially Low Battery State of Charge.

photovoltaic generation. When the SoC reaches the predefined lower limit of 50 %, the backup source automatically engages, ensuring uninterrupted service (transition time approximately 20 ms), and continues to supply the load. This transition occurs precisely at $t = 00:48$, as depicted by the temporal evolution of the SoC in Fig. 25.b. This configuration is maintained until photovoltaic power generation matches the load demand, occurring around $t = 01:13$. Subsequently, when PV production exceeds consumption, the battery enters a recharging phase, and energy injection into the grid is observed within the interval [01:52–02:10], whenever the surplus surpasses the acceptable charging power for the energy storage system.

The final scenario presented in Fig. 26 aims to validate the microgrid system's autonomy under unfavorable conditions. To achieve this, a low-irradiance profile with intermittent shading periods was imposed.

Analysis of the results indicates that the load is initially supplied by the energy storage system due to the absence of photovoltaic generation. Subsequently, combined power delivery from the main source (PV) and the supplementary source (battery) occurs once solar generation becomes partially available, though insufficient to meet the total demand.

Fig. 26.b illustrates the progressive discharge of the battery, whose state-of-charge decreases from 98 % to 58 %, followed by a brief recharging phase within the interval [01:57–02:09], partially restoring

the SoC to approximately 60 %.

The peaks observed in the experimental figures are mainly attributable to three instrumental factors. First, the Fronius inverter relies on a hybrid MPPT strategy that combines continuous Perturb and Observe tracking with short Particle Swarm Optimization sweeps triggered every ten minutes, and these 1 to 2 s sweeps can induce impulsive variations in PV power. Second, in the absence of supercapacitors that would normally buffer pump start up transients, the backup source is used to supply rapid current demands in order to protect the storage system, which is clearly visible in Fig. 26.a at 00:45 and 01:57. Third, the ITECH emulator is limited to 150 samples per irradiance temperature profile and operates with a sample and hold scheme that does not constrain the inter sample rate of change, generating near instantaneous discontinuities and step like energy flow curves, as observed in Fig. 26.a at 01:18, 01:35, and 01:43 when irradiance abruptly shifts between low and high levels. The presence or absence of these peaks in the recorded data closely depends on the temporal resolution of the data acquisition system, whose sampling interval is fixed at 5 s through communication between Node-RED and Google Sheets.

Beyond these measurement related effects, the scenario in Fig. 26 reveals an additional limitation of lead acid storage compared with lithium-ion technologies. Although the sizing aimed to ensure daily

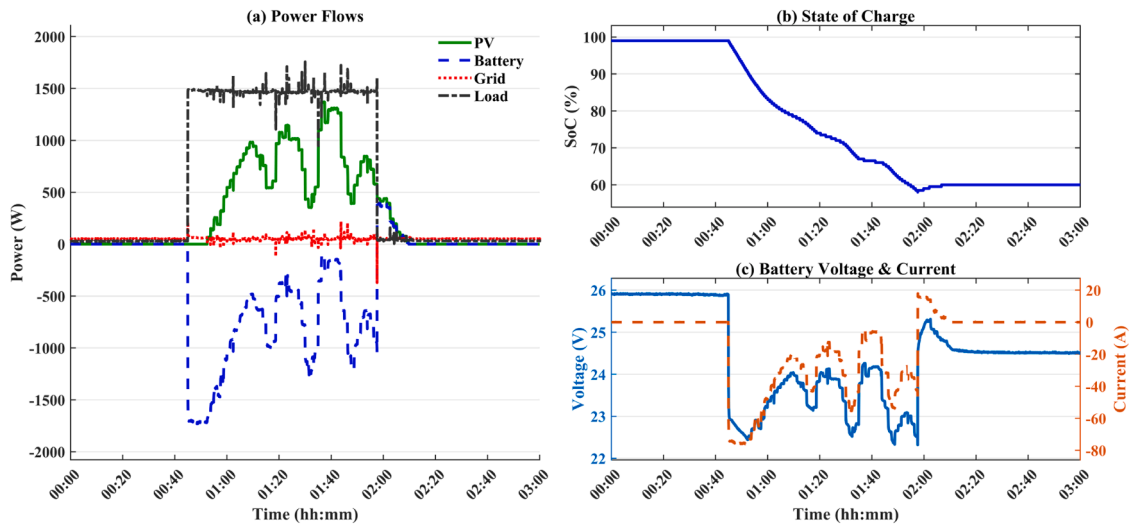


Fig. 26. Experimental Results – Scenario 3: Low-Irradiance Day.

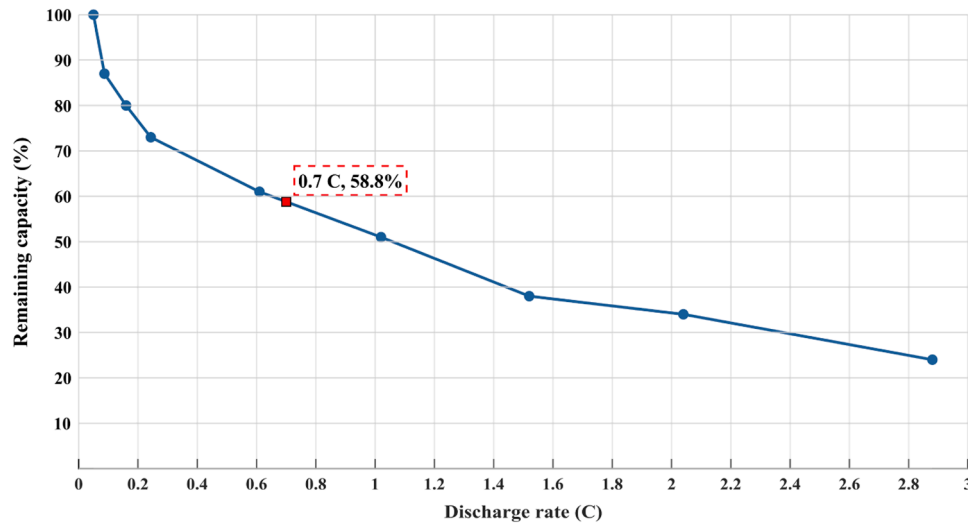


Fig. 27. Relationship Between Usable Capacity of a GEL Battery and Discharge Current [54].

autonomy under accelerated conditions, experiments show a pronounced SoC decrease reaching critical values despite partial PV support. This discrepancy is explained by the dependence of effective battery capacity on discharge current, since nominal capacity is typically specified at low rates such as C/20 and decreases significantly as current increases. In the laboratory tests, discharge currents reached up to 0.7 C as reported in Fig. 26.c, which, according to manufacturer specifications, can reduce usable capacity by up to 40 % during these periods [54]. This rate sensitivity complicates lead acid sizing under variable operating scenarios, thereby constituting a practical limitation of lead-acid storage under sustained high-discharge current conditions. The following figure illustrates the relationship between discharge current and recoverable capacity. Fig. 27.

In summary, the experimental tests carried out on the laboratory-scale prototype with reduced power capacity validated the overall coherence of the energy management sequences, while highlighting certain limitations inherent to the employed technologies. Additionally, the sensitivity of lead-acid batteries to discharge currents underscores the advantage of adopting more robust storage technologies to ensure autonomy under dynamic operating conditions.

5. Conclusion & perspectives

This study contributes to the sustainable modernization of Moroccan rural agriculture by presenting the design, optimal sizing, and experimental validation of a microgrid system.

Initially, the study provided a precise characterization of the farm's energy requirements. The full-scale installation intended for the RIM-SAR agricultural site (a partner of the MG-FARM project) was defined using PVSyst software, resulting in an optimal photovoltaic capacity of 25 kWp and a storage capacity of 2,828 Ah at 51 V. Subsequently, a comparative analysis between AC- and DC-coupling topologies was conducted in order to choose the converters design. This analysis demonstrated the superiority of the AC architecture in the studied context, yielding an efficiency gain of up to 2.1 %.

Building on this architectural choice, the converters were dimensioned as follows:

- For the photovoltaic inverter: A techno-economic optimization using a Pareto front analysis identified an optimal sizing ratio of $RS_{knee} = 1.0$. This ratio represents a balanced trade-off between

maximizing injected energy and minimizing the Levelized Cost of Electricity (LCOE), corresponding to a 25-kW on-grid inverter.

- For the bidirectional inverter-charger: Following industrial recommendations to match the nominal power of the PV inverter, a capacity of 25 kVA was selected for the farm-scale microgrid.

Furthermore, a 25-year economic and environmental assessment substantiated the advantages of upgrading the existing installation to the proposed microgrid system, highlighting estimated cost savings of €26,000 and a 98 % reduction in CO₂ emissions.

To validate these findings, a reduced-scale laboratory version of the microgrid was designed. This prototype relied on a theoretical source sizing approach specifically adapted to the constraints of accelerated-time testing. This configuration comprised a 2 kW PV system and a 120 Ah storage unit at 24 V. The power converters followed the same sizing logic as the full-scale system, resulting in a 2 kW photovoltaic inverter and a 2 kVA bidirectional converter.

Finally, experimental results obtained from the laboratory prototype under accelerated-time conditions demonstrated continuous load supply without interruption. The operating sequences respected both operational and technological constraints, confirming the robustness of the proposed design and the consistency of the implemented energy management strategy.

This work provides a concrete contribution to the energy transition of agricultural systems in rural Morocco by proposing a transferable methodology for deploying microgrids adapted to African contexts. Future perspectives are structured around three main directions:

1. The integration of advanced energy management techniques, particularly those based on artificial intelligence, to enable real-time predictive optimization;
2. The extension of the concept to neighboring farms through the deployment of interconnected microgrids, forming a collaborative and resilient architecture;
3. The exploration of new renewable energy sources, such as green hydrogen, to enhance the autonomy and flexibility of the overall system.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to improve the clarity and readability of the text. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

CRediT authorship contribution statement

Ayoub Rahmouni: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Driss Yousfi:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Amine El Houre:** Writing – review & editing, Visualization, Validation. **Mohamed Bakhouya:** Writing – review & editing, Supervision, Project administration, Data curation. **Mohammed Chaker:** Writing – review & editing, Resources. **Mohammed Bachiri:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Driss Yousfi reports financial support, administrative support, and equipment, drugs, or supplies were provided by Morocco Ministry of Higher Education Scientific Research and Innovation. If there are other authors, they declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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